

Project Report: Todo App Full Stack - Full Stack Web Developer

Written by Gagan Pasupuleti

Family:	Full Stack Web Developer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

React frontend + Express API + SQLite: create, complete, delete todos with persistence.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

Todo App Full Stack** is a beginner-level end-to-end project for Full Stack Web Developer. **Duration:** 2 weeks. **React frontend + Express API + SQLite:** create, complete, delete todos with persistence. **Tools:** React, Node.js, Express, SQLite.

Code Reference:

```
Objective: REST API for todos
Objective: React UI with forms
Objective: SQLite persistence
Objective: Basic error handling
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Full Stack Web Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Full Stack Web Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: API [Beginner]

Chapter focus: Phase: API** *(Level: Beginner)

Express CRUD routes. Complete these tasks: SQLite schema; Validation. Deliverable for this phase: API tested with curl.

Code Reference:

- [] SQLite schema
- [] Validation

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase API is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the API tested with curl to the teacher.

Chapter 3: Phase: UI [Beginner]

Chapter focus: Phase: UI** *(Level: Beginner)

React todo list. Complete these tasks: Add/complete/delete; Optimistic UI. Deliverable for this phase: Working frontend.

Code Reference:

- [] Add/complete/delete
- [] Optimistic UI

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase UI is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Working frontend to the teacher.

Chapter 4: Phase: Connect [Beginner]**Chapter focus: Phase: Connect** *(Level: Beginner)**

fetch from React. Complete these tasks: CORS setup; Loading states. Deliverable for this phase: Full stack running.

Code Reference:

- [] CORS setup
- [] Loading states

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Connect is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Full stack running to the teacher.

Chapter 5: Phase: Deploy [Beginner]**Chapter focus: Phase: Deploy** *(Level: Beginner)**

Frontend + API deploy. Complete these tasks: Vercel + Render; Env vars. Deliverable for this phase: Live URLs.

Code Reference:

- [] Vercel + Render
- [] Env vars

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Deploy is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Live URLs to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: GitHub monorepo; Live frontend; Live API. Skills demonstrated: React, Express, SQL. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: GitHub monorepo
Deliverable: Live frontend
Deliverable: Live API
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits GitHub monorepo and passes the rubric.